N6 source correction and weighted-to-affine coefficient bridge

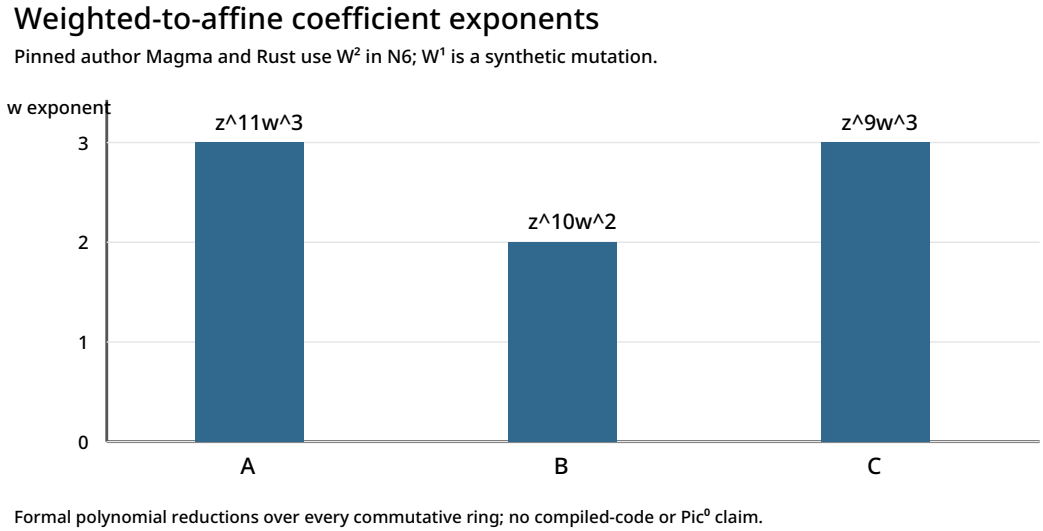
Source-bound review packet

25 October 2026

The pinned author Magma file at commit `ec98e115ed359c8b7facf559eb0d9f240703f6f5` assigns `WW1:=W1^2` before multiplying `N6` by `WW1`. The Rust runtime likewise calls its weighted doubling variant with the squared-w flag. The earlier claim of an author-code W^1 discrepancy was an alias-reading error. The retained $p = 739$, $W = 3$ mismatch is a synthetic W^1 mutation control. Raw certificates, field names, and collector cost rows are unchanged.

The published article defines $(Q, R, S, T, Z, W) = (qz^2, rz^4, sz^3w, tz^5w, z, w)$ and prints affine doubling coefficients A, B, C in Eq. (7). The five new Lean theorems prove the following polynomial equalities over every commutative ring:

coefficient	pinned W^2 scaling	synthetic W^1 defect
A	$z^{11}w^3A_{\text{affine}}$	$z^{11}w^2(w-1)rs(2r-3q^2-a_3)$
B	$z^{10}w^2B_{\text{affine}}$	0
C	$z^9w^3C_{\text{affine}}$	$z^9w^2(w-1)t(2r-3q^2-a_3)$



The proof log lists only `propext` and `Quot.sound`. The validator checks the full pinned author-file SHA-256 `7416a7fffaafff371738e7498a6795057372e1c477e0079297eebf2347c2e72a` and Rust SHA-256 `5976e7f30b3e9c5c49a5fc0f4095404b7322832e4b02f467bc3cb7e0d9dcc8c0`. This is a coefficient-level source and polynomial bridge; it does not prove complete Magma/Rust equivalence, compiled-code semantics, or the smooth-projective/ Pic^0 class law. See the CSV, Lean source, log, and validator beside this PDF.

Sources: pinned author Magma file; published article.